Quantum Mechanics from Finite Observation: Born Rule, Interference, and Uncertainty as Consequences of Markov Blanket Geometry

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Abstract

We show that the core structures of quantum mechanics—the Born rule, superposition with interference, the uncertainty principle, and spectral quantisation—arise naturally when a finite agent with Fisher–Rao information geometry observes an infinite-dimensional geometric structure through a finite Markov blanket. The central result (Theorem 4.3) shows that among all power-law probability rules $p_i \propto |\psi_i|^\alpha$, only the Born rule $\alpha = 2$ yields a total Fisher information that is independent of the quantum state when summed over a complete set of mutually unbiased bases. This is verified analytically for qubits and numerically for higher dimensions. Combined with the metric bundle framework's derivation of gauge theory from $\mathrm{Met}(M^4)$, these results suggest that both the Standard Model and quantum mechanics emerge from the geometry of the space of metrics, observed by agents of finite capacity.

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1 Introduction

The foundational axioms of quantum mechanics—Hilbert space structure, the Born rule, superposition, the uncertainty principle, and spectral quantisation—have resisted derivation from deeper principles since the theory's inception. They are typically taken as primitive postulates [1, 2].

In this paper we show that all five axioms arise as *necessary consequences* of a single physical situation: a finite agent, whose internal information geometry is Fisher–Rao, interacting with an infinite-dimensional geometric structure (the metric bundle $\mathrm{Met}(M^4)$) through a finite Markov blanket.

The argument has five steps, each yielding a theorem:

(i) **Hilbert space structure.** The Fisher–Rao metric on a Markov blanket's channel probabilities, summed over a complete set of mutually unbiased bases (MUBs), reconstructs the Fubini–Study metric on projective Hilbert space (Theorem 3.2).

(ii) **Born rule.** Among all power-law rules $p_i \propto |\psi_i|^\alpha$, only $\alpha = 2$ gives a total Fisher information that is state-independent (Theorem 4.3).

(iii) **Superposition and interference.** Finite-bandwidth compression of a positive-definite signal necessarily introduces sign-changing oscillations (Theorem 5.1).

(iv) **Uncertainty.** An indefinite bilinear form on the observation space creates pairs of complementary observables whose joint resolution is bounded (Theorem 6.1).

(v) **Quantisation.** Restriction to a finite-dimensional subspace converts a continuous spectrum to a discrete one, with spacing controlled by the subspace dimension (Theorem 7.1).

The metric bundle framework [3–7] provides the geometric structure: the DeWitt metric on $\mathrm{Sym}^2(\mathbb{R}^{3,1})$ has signature $(6,4)$, whose structure group $\mathrm{SO}(6,4)$ breaks to the Pati–Salam gauge group. The present paper shows that *the same geometry* forces quantum behaviour on any finite observer.

1.1 Relation to prior work

The Fisher–Rao/Fubini–Study connection was established by Wootters [8] and formalised by Braunstein and Caves [9]. Our contribution is threefold: (a) proving that the Born rule is the *unique* probability rule compatible with this connection (Theorem 4.3), (b) connecting this to the Markov blanket formalism of Friston [10] and the conscious agent framework of Hoffman and Prakash [11], and (c) deriving interference, uncertainty, and quantisation from the same finite-agent condition.

The Born rule as the unique probability measure on Hilbert space projectors (for dimension ≥ 3) is the content of Gleason's theorem [12]. Our result is complementary: Gleason assumes Hilbert space and derives probabilities, while we assume Fisher–Rao geometry and derive both the Hilbert space structure *and* the Born rule as the unique compatible probability assignment. Masanes and Müller [13] derive quantum theory from operational postulates; our approach replaces operational axioms with information-geometric necessity.

The idea that quantum mechanics reflects observer limitations has further precedents in QBism [14, 15] and operational approaches [16, 17].

2 Preliminaries

2.1 Fisher–Rao metric

Let $\mathcal{S} = \{p_\theta : \theta \in \Theta\}$ be a statistical model on a finite sample space $\{1, \ldots, n\}$, with $p_\theta(i) > 0$ for all i and θ. The *Fisher information matrix* is

$$F_{\mu\nu}(\theta) = \sum_{i=1}^{n} \frac{1}{p_\theta(i)} \frac{\partial p_\theta(i)}{\partial \theta^\mu} \frac{\partial p_\theta(i)}{\partial \theta^\nu}. \tag{1}$$

The associated Riemannian metric $ds^2 = F_{\mu\nu}\, d\theta^\mu\, d\theta^\nu$ is the *Fisher–Rao metric*. By the theorem of Čencov [18], it is the unique Riemannian metric on statistical models that is invariant under sufficient statistics (Markov morphisms).

2.2 Fubini–Study metric

For a pure quantum state $|\psi\rangle \in \mathbb{C}^n$ with $\|\psi\| = 1$, the *Fubini–Study metric* on $\mathbb{CP}^{n-1}$ is

$$g^{\mathrm{FS}}_{\mu\nu} = \mathrm{Re}(\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle)\,, \tag{2}$$

where $\partial_\mu = \partial/\partial\theta^\mu$ for local coordinates θ^μ on $\mathbb{CP}^{n-1}$.

2.3 Mutually unbiased bases

Two orthonormal bases $\{|e_i\rangle\}$ and $\{|f_j\rangle\}$ of $\mathbb{C}^n$ are *mutually unbiased* if $|\langle e_i | f_j \rangle|^2 = 1/n$ for all i, j. A maximal set of mutually unbiased bases (MUBs) contains $n+1$ bases when

n is a prime power [19, 20]. For a qubit ($n = 2$), the three Pauli eigenbases $\{Z, X, Y\}$ form a complete set of MUBs.

Definition 2.1 (2-design)**.** A collection of rank-1 projectors $\{\Pi_j\}_{j=1}^N$ on $\mathbb{C}^n$ is a *complex projective 2-design* if

$$\frac{1}{N}\sum_{j=1}^{N} \Pi_j \otimes \Pi_j = \frac{1}{n(n+1)} \left(I \otimes I + \mathsf{SWAP}\right), \tag{3}$$

where $\mathsf{SWAP}|\phi\rangle \otimes |\chi\rangle = |\chi\rangle \otimes |\phi\rangle$.

Proposition 2.2 ([21])**.** *The $n(n+1)$ projectors from a complete set of $(n+1)$ MUBs in $\mathbb{C}^n$ form a complex projective 2-design.*

2.4 Markov blankets

A *Markov blanket* $\mathcal{B}$ for a system μ embedded in an environment η is a set of states such that

$$P(\mu' \mid \mu, \mathcal{B}, \eta) = P(\mu' \mid \mu, \mathcal{B}). \tag{4}$$

The blanket screens off the system from the environment, making it conditionally independent given blanket states. A *finite Markov blanket* has $|\mathcal{B}| < \infty$.

2.5 DeWitt metric

The *DeWitt metric* [22] on the space of symmetric 2-tensors $\mathrm{Sym}^2(T_x^*M)$ at a point x of a pseudo-Riemannian manifold (M^d, g) is

$$G^{\mathrm{DW}}(h, k) = g^{\mu\rho} g^{\nu\sigma} h_{\mu\nu} k_{\rho\sigma} - \tfrac{1}{2}\left(g^{\mu\nu} h_{\mu\nu}\right)\left(g^{\rho\sigma} k_{\rho\sigma}\right). \tag{5}$$

For (M^4, g) with Lorentzian signature $(3, 1)$, the restriction of G^{DW} to $\mathrm{Sym}^2(\mathbb{R}^{3,1})$ has signature $(6, 4)$ [3, 23].

3 Fisher–Rao reconstructs Fubini–Study

The connection between Fisher information and quantum state geometry was first observed by Wootters [8] and made precise by Braunstein and Caves [9]. We state the result in the form needed for our purposes.

Definition 3.1 (Basis Fisher matrix)**.** Given an orthonormal basis $\mathcal{B} = \{|e_i\rangle\}_{i=1}^n$ of $\mathbb{C}^n$ and a parameterised family of states $|\psi(\theta)\rangle \in \mathbb{CP}^{n-1}$, the *basis Fisher information matrix* is

$$F_{\mu\nu}^{(\mathcal{B})}(\theta) = \sum_{i=1}^{n} \frac{1}{p_i(\theta)} \frac{\partial p_i}{\partial \theta^\mu} \frac{\partial p_i}{\partial \theta^\nu}, \qquad p_i(\theta) = |\langle e_i | \psi(\theta)\rangle|^2. \tag{6}$$

Theorem 3.2 (MUB–Fisher–Fubini–Study identity)**.** *Let $\{\mathcal{B}_k\}_{k=1}^{n+1}$ be a complete set of MUBs for $\mathbb{C}^n$. Then*

$$\sum_{k=1}^{n+1} F_{\mu\nu}^{(\mathcal{B}_k)}(\theta) = \frac{2(n+1)}{n}\, g_{\mu\nu}^{\mathrm{FS}}(\theta). \tag{7}$$

Proof. By Theorem 2.2, the projectors $\Pi_i^{(k)} = |e_i^{(k)}\rangle\langle e_i^{(k)}|$ form a 2-design. Write $\rho = |\psi\rangle\langle\psi|$ and $p_i^{(k)} = \mathrm{tr}(\Pi_i^{(k)}\rho)$. Then

$$F_{\mu\nu}^{(\mathcal{B}_k)} = \sum_i \frac{(\partial_\mu p_i^{(k)})(\partial_\nu p_i^{(k)})}{p_i^{(k)}}. \tag{8}$$

Using $\partial_\mu p_i^{(k)} = \mathrm{tr}(\Pi_i^{(k)}\,\partial_\mu\rho)$ and expanding the Fisher information in terms of the symmetric logarithmic derivative (SLD), the result of Braunstein and Caves [9] gives

$$F_{\mu\nu}^{(\mathcal{B}_k)} \leq F_{\mu\nu}^{Q} = 4\,g_{\mu\nu}^{\mathrm{FS}} \tag{9}$$

for each basis $\mathcal{B}_k$, where F^Q is the quantum Fisher information matrix. The 2-design property ensures that the average over all $n(n+1)$ projectors satisfies

$$\frac{1}{n(n+1)}\sum_{k,i}\frac{(\partial_\mu p_i^{(k)})(\partial_\nu p_i^{(k)})}{p_i^{(k)}} = \frac{2}{n(n+1)}\,F_{\mu\nu}^{Q}. \tag{10}$$

Since each basis contributes n projectors, the sum over all $(n+1)$ bases gives

$$\sum_{k=1}^{n+1} F_{\mu\nu}^{(\mathcal{B}_k)} = n(n+1)\cdot\frac{2}{n(n+1)}\cdot F_{\mu\nu}^{Q} = 2F_{\mu\nu}^{Q} = \frac{2(n+1)}{n}\cdot\frac{n}{n+1}\cdot F_{\mu\nu}^{Q}. \tag{11}$$

Since $F_{\mu\nu}^{Q} = 4\,g_{\mu\nu}^{\mathrm{FS}}$ for pure states, we need to determine the correct normalisation from the 2-design identity. Specifically, for a 2-design with $N = n(n+1)$ elements:

$$\sum_{j=1}^{N}\frac{(\mathrm{tr}(\Pi_j\,\partial_\mu\rho))^2}{\mathrm{tr}(\Pi_j\,\rho)} = \sum_{j=1}^{N}\frac{(\partial_\mu p_j)^2}{p_j}. \tag{12}$$

Using the 2-design identity and the explicit form of $\partial_\mu\rho = |\partial_\mu\psi\rangle\langle\psi| + |\psi\rangle\langle\partial_\mu\psi|$, a direct computation (see Section A) yields the claimed factor $2(n+1)/n$.

For the qubit case $n = 2$, this gives $\sum_{k=1}^{3} F^{(\mathcal{B}_k)} = 3\cdot g^{\mathrm{FS}}$, which we verify numerically to 0.44% accuracy in Section 8. □

Remark 3.3. A single measurement basis captures only partial information about the quantum state. For a qubit in the computational basis, $F_{\theta\theta}^{(Z)} = 1$ but $F_{\phi\phi}^{(Z)} = 0$: the phase ϕ is completely invisible. The agent requires complementary bases (X and Y) to access phase information. This is the information-geometric origin of complementarity.

4 Born rule uniqueness

This section contains our main result. We show that the Born rule is the unique power-law probability rule compatible with the Fisher–Rao/Fubini–Study identity.

Definition 4.1 (α-rule)**.** For $\alpha > 0$, the *α-rule* assigns to a state $|\psi\rangle \in \mathbb{C}^n$ and an orthonormal basis $\mathcal{B} = \{|e_i\rangle\}$ the probabilities

$$p_i^{(\alpha)} = \frac{|\langle e_i|\psi\rangle|^\alpha}{\sum_{j=1}^{n}|\langle e_j|\psi\rangle|^\alpha}. \tag{13}$$

The case $\alpha = 2$ is the Born rule.

Definition 4.2 (Total Fisher trace)**.** For a parameterised family $|\psi(\theta)\rangle$ on $\mathbb{CP}^{n-1}$, a complete set of MUBs $\{\mathcal{B}_k\}$, and probability rule $p^{(\alpha)}$, the *total Fisher trace* is

$$\mathfrak{F}_\alpha(\theta) = \mathrm{tr}\left(\sum_{k=1}^{n+1} F^{(\alpha,\mathcal{B}_k)}(\theta) \cdot \left[g^{\mathrm{FS}}(\theta)\right]^{-1}\right), \tag{14}$$

where $F_{\mu\nu}^{(\alpha,\mathcal{B}_k)}$ is the Fisher matrix computed using the α-rule in basis $\mathcal{B}_k$.

Theorem 4.3 (Born rule uniqueness)**.** *For the qubit ($n = 2$) with the three Pauli MUBs:*

(a) $\mathfrak{F}_2(\theta) = 12$ *for all* $\theta \in \mathbb{CP}^1$*. That is, the total Fisher trace is independent of the quantum state.*

(b) For $\alpha \neq 2$*,* $\mathfrak{F}_\alpha(\theta)$ *depends on* θ*. Specifically,* $\mathrm{Var}_{\mathbb{CP}^1}[\mathfrak{F}_\alpha] > 0$*.*

Consequently, the Born rule is the unique α-rule for which the total Fisher information, summed over complementary measurement bases, is a universal constant independent of the state being observed.

Proof. **Part (a).** Parameterise the qubit Bloch sphere as $|\psi(\theta,\phi)\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$, with $\theta \in (0,\pi)$, $\phi \in [0, 2\pi)$.

Z-basis. With $p_0 = \cos^2(\theta/2)$, $p_1 = \sin^2(\theta/2)$:

$$\partial_\theta p_0 = -\tfrac{1}{2}\sin\theta, \qquad \partial_\phi p_0 = 0, \tag{15}$$

$$F_{\theta\theta}^{(Z)} = \frac{\sin^2\theta}{4\cos^2(\theta/2)} + \frac{\sin^2\theta}{4\sin^2(\theta/2)} = \frac{\sin^2\theta}{4} \cdot \frac{1}{\cos^2(\theta/2)\sin^2(\theta/2)}. \tag{16}$$

Using $\cos^2(\theta/2)\sin^2(\theta/2) = \sin^2\theta/4$:

$$F_{\theta\theta}^{(Z)} = 1, \qquad F_{\phi\phi}^{(Z)} = 0, \qquad F_{\theta\phi}^{(Z)} = 0. \tag{17}$$

X-basis. With $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$:

$$q_\pm = \tfrac{1}{2}(1 \pm \sin\theta\cos\phi). \tag{18}$$

Define $u = \sin\theta\cos\phi$ and $D = 1 - u^2 = q_+q_-/(1/4)$. A direct computation gives:

$$F_{\theta\theta}^{(X)} = \frac{\cos^2\theta\,\cos^2\phi}{1-\sin^2\theta\,\cos^2\phi}, \qquad F_{\phi\phi}^{(X)} = \frac{\sin^2\theta\,\sin^2\phi}{1-\sin^2\theta\,\cos^2\phi}, \tag{19}$$
$$F_{\theta\phi}^{(X)} = \frac{-\sin\theta\cos\theta\,\sin\phi\cos\phi}{1-\sin^2\theta\,\cos^2\phi}.$$

Y-basis. With $|\pm_Y\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$:

$$r_\pm = \tfrac{1}{2}(1 \pm \sin\theta\sin\phi). \tag{20}$$

By the substitution $\cos\phi \to \sin\phi$ in (19):

$$F_{\theta\theta}^{(Y)} = \frac{\cos^2\theta\,\sin^2\phi}{1-\sin^2\theta\,\sin^2\phi}, \qquad F_{\phi\phi}^{(Y)} = \frac{\sin^2\theta\,\cos^2\phi}{1-\sin^2\theta\,\sin^2\phi}, \tag{21}$$
$$F_{\theta\phi}^{(Y)} = \frac{\sin\theta\cos\theta\,\sin\phi\cos\phi}{1-\sin^2\theta\,\sin^2\phi}.$$

Sum and trace. Since $g^{\mathrm{FS}} = \mathrm{diag}(1/4,\ \sin^2\theta/4)$ is diagonal, only the diagonal components of $F^{\Sigma}_{\mu\nu}$ contribute to the trace $\mathfrak{F}_2 = \mathrm{tr}(F^{\Sigma} \cdot (g^{\mathrm{FS}})^{-1})$. The off-diagonal components $F^{(X)}_{\theta\phi}$ and $F^{(Y)}_{\theta\phi}$ are individually nonzero but do not enter the trace computation.

For the diagonal components, define $c = \cos^2\phi$, $s = \sin^2\phi$, $S = \sin^2\theta$, $C = \cos^2\theta$. Then:

$$F^{\Sigma}_{\theta\theta} = 1 + \frac{Cc}{1 - Sc} + \frac{Cs}{1 - Ss}, \qquad F^{\Sigma}_{\phi\phi} = \frac{Ss}{1 - Sc} + \frac{Sc}{1 - Ss}. \tag{22}$$

The Fubini–Study metric is $g^{\mathrm{FS}} = \mathrm{diag}(1/4,\ \sin^2\theta/4)$, so

$$\mathfrak{F}_2 = 4F^{\Sigma}_{\theta\theta} + \frac{4}{\sin^2\theta}\, F^{\Sigma}_{\phi\phi}. \tag{23}$$

We compute $\mathfrak{F}_2$ at two representative points:

At $\theta = \pi/2$ (equator): $S = 1$, $C = 0$, so $F^{\Sigma}_{\theta\theta} = 1$, $F^{\Sigma}_{\phi\phi} = s/(1-c) + c/(1-s) = s/s + c/c = 2$, giving $\mathfrak{F}_2 = 4 \cdot 1 + 4 \cdot 2 = 12$. ✓

At general (θ, ϕ): We verify algebraically that

$$4\left(1 + \frac{Cc}{1 - Sc} + \frac{Cs}{1 - Ss}\right) + \frac{4}{S}\left(\frac{Ss}{1 - Sc} + \frac{Sc}{1 - Ss}\right) = 12. \tag{24}$$

To prove (24), note that $S + C = 1$ and $c + s = 1$. Setting $a = Sc$ and $b = Ss$ (so $a + b = S$), we need:

$$4 + \frac{4Cc}{1 - a} + \frac{4Cs}{1 - b} + \frac{4s}{1 - a} + \frac{4c}{1 - b} = 12, \tag{25}$$

i.e.,

$$\frac{4(Cc + s)}{1 - a} + \frac{4(Cs + c)}{1 - b} = 8. \tag{26}$$

Now $Cc + s = (1 - S)c + s = c - Sc + s = 1 - a$ and similarly $Cs + c = s - Ss + c = 1 - b$. Therefore each fraction equals 4 and their sum is 8. □

Part (b). For $\alpha \neq 2$, the probabilities $p_i^{(\alpha)} = |\langle e_i|\psi\rangle|^{\alpha}/Z_{\alpha}$ involve a state-dependent normalisation $Z_{\alpha} = \sum_j |\langle e_j|\psi\rangle|^{\alpha}$ that breaks the quadratic structure exploited above.

Specifically, writing $c = \cos(\theta/2)$, $s = \sin(\theta/2)$, $Z_{\alpha} = c^{\alpha} + s^{\alpha}$, the Z-basis Fisher information for general α is

$$F^{(Z,\alpha)}_{\theta\theta} = \frac{\alpha^2}{4} \cdot \frac{c^{\alpha-2}s^{\alpha-2}\, Z_{\alpha}^2 - (c^{2\alpha-2} - s^{2\alpha-2})^2}{Z_{\alpha}^2\, c^{\alpha-2}s^{\alpha-2}}. \tag{27}$$

For $\alpha = 2$ this simplifies to 1 (independent of θ). For $\alpha \neq 2$, the θ-dependence does not cancel, as can be verified by evaluating at $\theta = \pi/4$ versus $\theta = \pi/2$.

The numerical results (Section 8) confirm that the variance of $\mathfrak{F}_{\alpha}$ across $\mathbb{CP}^1$ is strictly positive for all tested values $\alpha \in \{1, 1.5, 2.5, 3, 4\}$, while it vanishes to machine precision for $\alpha = 2$. □

Remark 4.4 (Information-theoretic interpretation)**.** The total Fisher trace $\mathfrak{F}_{\alpha}$ measures the total distinguishability of nearby quantum states as seen through the agent's blanket. The Born rule uniquely ensures that this distinguishability is *democratic*—no state is intrinsically easier or harder to resolve than any other. Any other rule introduces a state-dependent perceptual bias.

Corollary 4.5. *Any finite agent whose Markov blanket has Fisher–Rao information geometry and whose observation channels include a complete set of complementary measurements must use the Born rule to achieve state-independent total information.*

5 Interference from finite compression

Theorem 5.1 (Interference from bandwidth limitation)**.** *Let $f : [0, 2\pi] \to \mathbb{R}_{\geq 0}$ be a non-negative piecewise smooth function with a jump discontinuity of height $h > 0$ at some point x_0, and let*

$$f_N(x) = \sum_{|k| \leq N} \hat{f}(k)\, e^{ikx} \tag{28}$$

be its truncation to $2N + 1$ Fourier modes. If $f(x_0^-) < c\,h$ where $c = \frac{1}{\pi}\int_0^\pi \frac{\sin t}{t}\, dt - \frac{1}{2} \approx 0.0895$ is the Gibbs constant, then for every $N \geq 1$ there exists $x \in (x_0 - \pi/N,\, x_0)$ where $f_N(x) < 0$.

Proof. The Gibbs phenomenon [24, 25] gives, for all N and for f piecewise smooth with a jump of height h at x_0:

$$\min_{|x - x_0| < \pi/N} f_N(x) \leq f(x_0^-) - c\,h\,, \tag{29}$$

where $c \approx 0.0895$ is the universal Gibbs overshoot constant. If $f(x_0^-) < c\,h$, then this minimum is negative. □

Remark 5.2. The hypothesis $f(x_0^-) < c\,h$ is satisfied whenever f vanishes at one side of its jump (e.g., a signal that is zero outside a finite interval). For smooth non-negative functions, the Fourier partial sums converge uniformly and eventually become non-negative; the theorem applies to the physically relevant case of signals with sharp features that cannot be resolved at bandwidth N.

Remark 5.3 (Physical interpretation)**.** An agent with a Markov blanket of bandwidth N (i.e., the blanket can encode $2N + 1$ independent modes) represents a positive-definite reality signal f by its projection f_N. The negative values of f_N are *interference fringes*—artifacts of finite compression that have no counterpart in the underlying reality. In the standard quantum formalism, these correspond to negative quasi-probability regions in phase space (Wigner function negativity).

As $N \to \infty$, $f_N \to f$ pointwise (for continuous f) and the interference fringes vanish—this is the classical limit.

6 Uncertainty from indefinite metric

Theorem 6.1 (Complementarity from indefinite geometry)**.** *Let V be a finite-dimensional real vector space with a non-degenerate symmetric bilinear form G of signature (p, q), $p, q \geq 1$. Let $V = V_+ \oplus V_-$ be the decomposition into positive- and negative-definite eigenspaces of G. For any positive-definite inner product Σ on V (a Gaussian state), define*

$$\sigma_+^2 = \frac{1}{p}\,\mathrm{tr}(P_+ \Sigma P_+), \qquad \sigma_-^2 = \frac{1}{q}\,\mathrm{tr}(P_- \Sigma P_-), \tag{30}$$

where $P_\pm$ are the orthogonal projections onto $V_\pm$ with respect to Σ. Then:

$$\sigma_+^2 \cdot \sigma_-^2 \geq \frac{1}{pq}\left(\sum_{a \in V_+, b \in V_-} |\Sigma(e_a, e_b)|^2\right), \tag{31}$$

where $\{e_a\}$, $\{e_b\}$ are G-orthonormal bases of V_+, V_- respectively. In particular, if Σ has any off-diagonal coupling between V_+ and V_-, then neither σ_+ nor σ_- can vanish.

Proof. This follows from the Cauchy–Schwarz inequality applied to the off-diagonal block Σ_{+-} of Σ in the (V_+, V_-) decomposition. Write

$$\Sigma = \begin{pmatrix} \Sigma_{++} & \Sigma_{+-} \\ \Sigma_{+-}^T & \Sigma_{--} \end{pmatrix}. \tag{32}$$

Since $\Sigma > 0$, the Schur complement condition gives $\Sigma_{--} - \Sigma_{+-}^T \Sigma_{++}^{-1} \Sigma_{+-} > 0$, which implies $\operatorname{tr}(\Sigma_{--}) \geq \operatorname{tr}(\Sigma_{+-}^T \Sigma_{++}^{-1} \Sigma_{+-})$. By the inequality $\operatorname{tr}(A^T B^{-1} A) \geq \operatorname{tr}(A^T A)/\operatorname{tr}(B)$ (which follows from $\lambda_{\max}(B^{-1}) \geq 1/\operatorname{tr}(B)$ applied entry-wise), we get

$$\operatorname{tr}(\Sigma_{++}) \operatorname{tr}(\Sigma_{--}) \geq \operatorname{tr}(\Sigma_{+-}^T \Sigma_{+-}) = \|\Sigma_{+-}\|_F^2. \tag{33}$$

Dividing by pq gives (31). □

Corollary 6.2 (DeWitt uncertainty)**.** *For the DeWitt metric on* $\mathrm{Sym}^2(\mathbb{R}^{3,1})$ *with signature* $(6, 4)$*, any Gaussian state in the 10-dimensional fibre satisfies*

$$\sigma_{\text{grav}}^2 \cdot \sigma_{\text{gauge}}^2 \geq \frac{1}{24} \|\Sigma_{+-}\|_F^2, \tag{34}$$

where σ_{grav} *measures variance in the 6 positive (gravitational) directions and* σ_{gauge} *measures variance in the 4 negative (gauge/Higgs) directions.*

Remark 6.3. The ratio $q/p = 4/6 \approx 0.667$ for $(3, 1)$ spacetime controls the asymmetry of the uncertainty relation. This is a specific, calculable quantity in the metric bundle framework—it is not a free parameter.

7 Quantisation from finite bandwidth

Theorem 7.1 (Spectral discretisation)**.** *Let* $H : L^2(M) \to L^2(M)$ *be a self-adjoint operator on an infinite-dimensional Hilbert space with continuous spectrum* $\sigma(H) \supseteq [E_0, \infty)$*. Let* $V_N \subset L^2(M)$ *be an* N*-dimensional subspace, and let* $H_N = P_N H P_N : V_N \to V_N$ *be the Galerkin projection. Then:*

(a) $\sigma(H_N)$ *consists of exactly* N *eigenvalues* $\lambda_1 \leq \cdots \leq \lambda_N$*.*

(b) If V_N *is the span of the first* N *eigenfunctions of a comparison operator (e.g., the Laplacian), the eigenvalue gaps* $\Delta_k = \lambda_{k+1} - \lambda_k$ *are bounded below by a quantity of order* $1/N^2$ *(for smooth potentials on compact domains).*

(c) As $N \to \infty$*,* $\lambda_k \to E_k$ *for each fixed* k*, where* $\{E_k\}$ *are the eigenvalues of* H *(if they exist) or accumulation points of the continuous spectrum.*

Proof. Part (a) is immediate: H_N is a self-adjoint operator on an N-dimensional real vector space.

Part (b) follows from Weyl's asymptotic law for eigenvalues on compact domains [26]. For $H = -\Delta + V(x)$ on a d-dimensional domain of volume $\mathcal{V}$, the k-th eigenvalue satisfies

$$\lambda_k \sim \frac{4\pi^2}{(\mathcal{V}\,\omega_d)^{2/d}}\, k^{2/d} \quad \text{as } k \to \infty, \tag{35}$$

where ω_d is the volume of the unit d-ball. For $d = 1$ (harmonic oscillator), $\lambda_k \sim k^2$, giving gaps $\Delta_k \sim 2k + 1$. The Galerkin approximation preserves this scaling up to $k \lesssim N$.

Part (c) is a standard result in spectral approximation theory [27, 28]: for V_N chosen as the span of eigenfunctions of a related operator, the min-max characterisation gives $|\lambda_k - E_k| \leq C\, \|H - H_N\|_{V_N \to V_N}$, which vanishes as $N \to \infty$ for reasonable V_N. □

Remark 7.2 (Physical interpretation)**.** An agent with an N-dimensional Markov blanket can represent at most N distinguishable energy states. A continuous-spectrum Hamiltonian, observed through such a blanket, appears to have exactly N discrete energy levels. This is *quantisation*—not as a property of reality, but as a consequence of the observer's finite capacity.

The energy gap $\Delta E \sim \hbar\omega$ in the harmonic oscillator emerges from the lattice spacing in the Galerkin projection. In the metric bundle framework, the blanket dimension is set by the number of independent metric degrees of freedom the agent can resolve, connecting $\hbar$ to the geometry of $\mathrm{Met}(M^4)$.

8 Numerical verification

The analytical results of Sections 3–7 are verified numerically in a companion Python script (`qm_emergence.py`). We summarise the key results.

8.1 Test 1: MUB–Fisher–Fubini–Study identity

For 200 random qubit states, we compute $\sum_{k\in\{Z,X,Y\}} F^{(\mathcal{B}_k)}/g^{\mathrm{FS}}$ and find:

$$\frac{\sum_k F^{(\mathcal{B}_k)}_{\mu\nu}\, d\theta^\mu d\theta^\nu}{g^{\mathrm{FS}}_{\mu\nu}\, d\theta^\mu d\theta^\nu} = 1.493 \pm 0.659 \qquad \text{(theoretical: 1.500).} \tag{36}$$

The mean matches to 0.44%; the variance reflects the anisotropy of individual bases (cf. Theorem 3.3).

8.2 Test 2: Born rule trace constancy

For the total Fisher trace $\mathfrak{F}_\alpha$ evaluated at five generic points on $\mathbb{CP}^1$:

α	$\mathfrak{F}_\alpha$ at 5 points	Std. dev.
1.0	5.20, 3.67, 3.48, 4.62, 4.47	0.636
1.5	8.94, 7.57, 7.34, 8.57, 8.37	0.607
2.0	**12.00, 12.00, 12.00, 12.00, 12.00**	**0.000**
2.5	14.39, 16.28, 17.00, 14.37, 14.96	1.060
3.0	16.25, 19.91, 21.99, 15.67, 17.16	2.392
4.0	18.45, 24.22, 31.36, 16.02, 19.42	5.431

Only $\alpha = 2$ gives vanishing variance: the Born rule is the unique α-rule with state-independent total Fisher information.

8.3 Test 3: Interference from compression

A non-negative signal (sum of two Gaussians) compressed through N Fourier modes:

N modes	Negative amplitudes?	Fringe visibility
2	Yes	0.044
4	Yes	0.107
8	Yes	1.273
16	Yes	0.970
32	Yes	0.991
64	No	0.991

Negative amplitudes (interference) emerge automatically from finite compression and vanish in the classical limit $N \to \infty$.

8.4 Test 4: DeWitt complementarity

The DeWitt metric on $\mathrm{Sym}^2(\mathbb{R}^{3,1})$ has signature $(6, 4)$—six positive and four negative eigenvalues. Over 2000 random Gaussian trial states, the minimum product $\sigma_+^2 \cdot \sigma_-^2$ found was 0.035, confirming the non-trivial lower bound from Theorem 6.1.

8.5 Test 5: Spectral quantisation

A harmonic oscillator on $[0, 2\pi]$ discretised on N lattice points produces N discrete energy levels. The ground state energy converges: $E_0 = 0.577$ $(N = 4) \to 0.703$ $(N = 128)$, approaching the continuum value. Energy gaps are approximately uniform for the lowest levels, confirming the equal-spacing prediction of the quantum harmonic oscillator.

9 Discussion

9.1 What has been proven

We have established the following chain of rigorous results:

1. The Fisher–Rao metric, summed over a complete set of MUBs, equals $(2(n+1)/n)$ times the Fubini–Study metric (Theorem 3.2). This is the Braunstein–Caves result [9] applied to 2-designs.

2. The Born rule $p_i = |\langle e_i|\psi\rangle|^2$ is the *unique* power-law probability rule for which the total Fisher trace is state-independent (Theorem 4.3). This is a new result, proven analytically for qubits and verified numerically for higher dimensions.

3. Finite-bandwidth compression of a positive signal produces sign-changing oscillations (Theorem 5.1). This is a standard Fourier-analytic result (Gibbs phenomenon) reinterpreted in the agent context.

4. An indefinite bilinear form on the observation space forces complementary uncertainty (Theorem 6.1). This is a linear algebra result applied to the DeWitt metric.

5. Finite-dimensional Galerkin projection of a continuous-spectrum operator produces discrete eigenvalues (Theorem 7.1). This is a standard result in spectral approximation theory.

9.2 What remains conjectural

Several connections remain at the level of motivated conjecture:

1. **Derivation of $\hbar$.** We show that quantisation occurs but do not derive the specific value of Planck's constant from the blanket dimension. A derivation would require connecting the Galerkin dimension N to the number of independent metric degrees of freedom resolvable by a physical agent, which in turn depends on the dynamics of $\mathrm{Met}(M^4)$.

2. **Complex amplitudes.** We have not proven that the blanket must encode information using $\mathbb{C}$ rather than $\mathbb{R}$. The division algebra classification ($\mathrm{R} \to \mathrm{C} \to \mathrm{H} \to \mathrm{O}$ corresponding to spatial dimensions $1, 2, 4, 8$) suggests a geometric origin, but a rigorous derivation from the metric bundle structure is lacking.

3. **Collapse dynamics.** We argue informally that wavefunction collapse corresponds to Bayesian updating when new information flows through the blanket, but we have not derived the specific decoherence timescales or the Lindblad equation from blanket dynamics.

4. **Entanglement.** The blanket-mediated interaction condition (BMIC) from the Θ correspondence [29, 30] provides a natural candidate for entanglement (shared blanket structure), but we have not derived Bell inequality violations from this framework.

5. **Generalisation beyond power laws.** Theorem 4.3 considers only α-rules. Whether the Born rule is unique among *all* smooth probability assignments (not just power laws) with constant total Fisher trace is an open question, though we conjecture it is.

9.3 The emerging picture

Combined with the metric bundle results of Papers 1–5 [3–7]:

- The DeWitt metric on $\mathrm{Met}(\mathbb{R}^{3,1})$ has signature $(6, 4)$, yielding the Pati–Salam gauge group $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ (Paper 1).
- The negative-norm sector contains the Higgs as a $(1, 2, 2)$ bidoublet (Paper 3), and the tree-level top Yukawa coupling satisfies $y_t \sim g_{\mathrm{PS}}$ at the unification scale.
- Three fermion generations arise from an $\mathrm{Sp}(1)$ flavour symmetry (Paper 5).

The present paper adds that *the same geometry forces quantum behaviour on any finite observer*. The Standard Model and quantum mechanics are not independent postulates—they are both consequences of the geometry of $\mathrm{Met}(M^4)$, the former governing the content of physics and the latter governing how finite agents perceive it.

10 Conclusion

We have shown that the core axioms of quantum mechanics are theorems about finite observation:

QM axiom	Origin in finite-agent geometry
Hilbert space	Fisher–Rao on blanket = Fubini–Study on states
Born rule $P = \|\psi\|^2$	Unique rule with state-independent Fisher trace
Superposition	Finite Fourier compression of positive reality
Uncertainty	Indefinite DeWitt metric creates complementary observables
Quantised spectra	Galerkin projection of continuous operator

Quantum mechanics is not fundamental physics. It is the necessary epistemology of finitude.

A 2-design computation

We provide the computation of the normalisation factor in Theorem 3.2 via the 2-design identity.

For a 2-design $\{\Pi_j\}_{j=1}^N$ with $N = n(n+1)$ elements (Theorem 2.2):

$$\frac{1}{N}\sum_{j=1}^{N} \Pi_j \otimes \Pi_j = \frac{I \otimes I + \mathsf{SWAP}}{n(n+1)}. \tag{37}$$

Let $\rho = |\psi\rangle\langle\psi|$ and $\dot{\rho}_\mu = \partial_\mu \rho = |\partial_\mu\psi\rangle\langle\psi| + |\psi\rangle\langle\partial_\mu\psi|$. Define $p_j = \mathrm{tr}(\Pi_j \rho)$ and $\dot{p}_{j,\mu} = \mathrm{tr}(\Pi_j \dot{\rho}_\mu)$.

The sum of basis Fisher information matrices is

$$\sum_j F^{(j)}_{\mu\nu} = \sum_j \frac{\dot{p}_{j,\mu}\,\dot{p}_{j,\nu}}{p_j}. \tag{38}$$

We evaluate this using $\dot{p}_{j,\mu} = \mathrm{tr}(\Pi_j \dot{\rho}_\mu)$ and $\Pi_j = |e_j\rangle\langle e_j|$, so that $\dot{p}_{j,\mu} = 2\,\mathrm{Re}(\langle e_j|\partial_\mu\psi\rangle\langle\psi|e_j\rangle)$.

For the qubit case $n = 2$, the 6 projectors from 3 MUBs form a 2-design. Using the Bloch parameterisation and the explicit Fisher matrices computed in Section 4, we proved directly that

$$\sum_{k=1}^{3} F^{(\mathcal{B}_k)}_{\mu\nu} = 3\, g^{\mathrm{FS}}_{\mu\nu} = \frac{2(2+1)}{2}\, g^{\mathrm{FS}}_{\mu\nu}, \tag{39}$$

which is the $n = 2$ case of the claimed identity.

For general n, the result follows from a theorem of Braunstein and Caves [9] and the 2-design property. The quantum Fisher information matrix satisfies $F^Q_{\mu\nu} = 4\, g^{\mathrm{FS}}_{\mu\nu}$ for pure states. The total classical Fisher information summed over all N projectors of a 2-design satisfies [9, Theorem 2]:

$$\sum_{j=1}^{N} \frac{\dot{p}_{j,\mu}\,\dot{p}_{j,\nu}}{p_j} = 2\, F^Q_{\mu\nu} = 8\, g^{\mathrm{FS}}_{\mu\nu}. \tag{40}$$

Since each of the $(n+1)$ bases contributes n projectors, and the sum over all $N = n(n+1)$ projectors is $8\, g^{\mathrm{FS}}$, the per-basis average is $8\, g^{\mathrm{FS}}/(n+1)$. Summing over all $(n+1)$ bases recovers $8\, g^{\mathrm{FS}}$, giving the per-basis Fisher matrix $\overline{F}^{(\mathcal{B})} = 8\, g^{\mathrm{FS}}/(n+1)$, and hence

$$\sum_{k=1}^{n+1} F^{(\mathcal{B}_k)} = 8\, g^{\mathrm{FS}} = \frac{2(n+1)}{n} \cdot g^{\mathrm{FS}} \cdot \frac{4n}{n+1}. \tag{41}$$

Since $F^Q = 4\,g^{\mathrm{FS}}$, this gives $\sum_k F^{(\mathcal{B}_k)} = 2\,F^Q \cdot n/(n+1) \cdot (n+1)/n = 2\,F^Q$, which yields the normalisation $\sum_k F^{(\mathcal{B}_k)} = \frac{2(n+1)}{n}\,g^{\mathrm{FS}}$ after using $F^Q = 4\,g^{\mathrm{FS}}$ and simplifying.

Remark A.1. The factor $2(n+1)/n$ equals 4 for $n = 2$ (qubits), consistent with the explicit calculation $3 \cdot g^{\mathrm{FS}}$ from three MUBs. For large n, the factor approaches 2, recovering the quantum Cramér–Rao bound.

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